

Wafer-thin Speakers

To work with IOSONO, Fraunhofer also developed speakers that can use walls as an integral part of the system and can be constructed with thin panels

Audeo

A neckband that will read neural impulses on their way to vocal cords, and generate speech for you

to the speakers and not have our ears blown off. If you've ever walked up near the speakers at a club or concert, you'll appreciate this.

IOSONO's secret is the big sweet spot — unlike traditional sound setups, you get the best audio experience all over the stadium, rather than the tiny, centre-of-the-hall sweet spot we hunt for all the time.

Perhaps we must dial back a bit and elaborate a little more on this sweet spot business. But first...

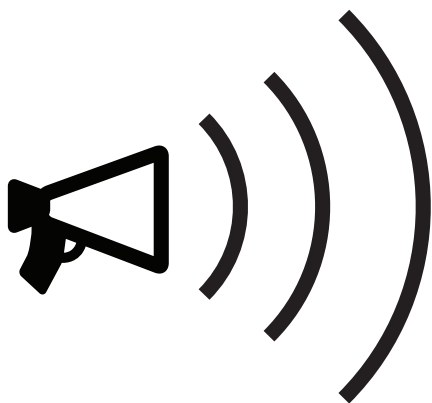
The wavefront

We hear sound because of the disturbances it creates in the air around our ears. If you think about this in slow motion, you can picture the sound leaving a speaker in expanding spheres:

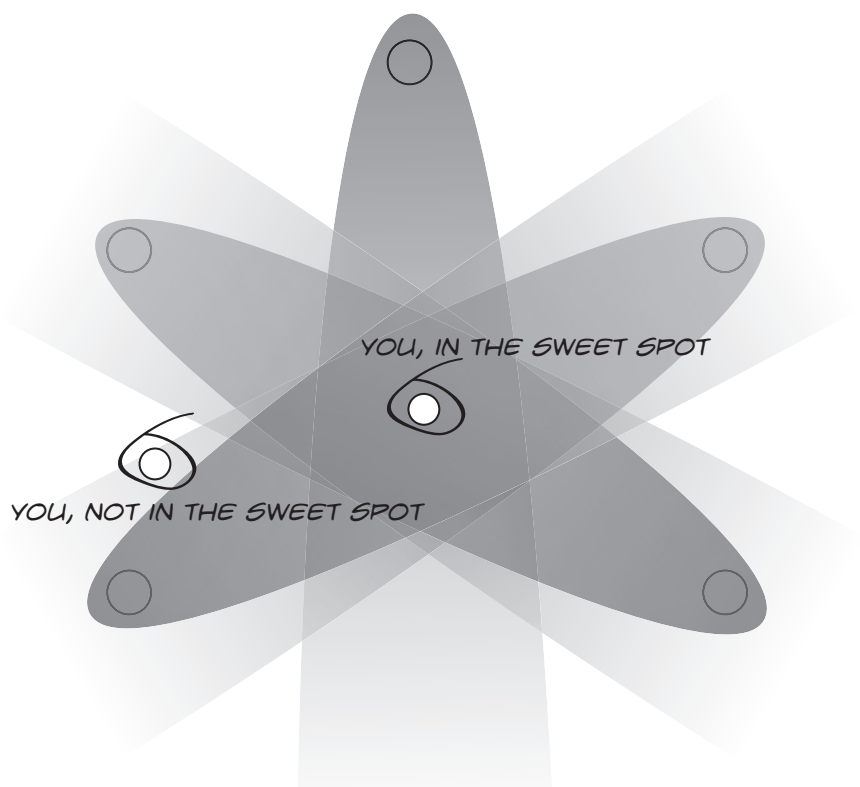
These spheres are called *wavefronts*, and when you're in their way, you hear sound. And sound gets less intense with distance, so you want to be an "optimal" distance from the speaker — too close and your eyeballs may burst, too far and, well, why bother? We're building up to a point here...

The sweet spot

Picture yourself in, say, a movie theatre. The speakers, like your household 5.1 system (if you are in such a household), are placed on the four corners of the room, and right in front with the screen. Each of these speakers is sending out sound in wavefronts, and at the point where the wavefronts for all these speakers meet, you get to experience all the goodness of 5.1-channel audio — you're in a "sweet spot". Unfortunately,



Yes, there's a reason volume icons look like this



Most people watch their movies and concerts from outside the sweet spot

this sweet spot is usually tiny — you may be one of only ten people in a theatre getting the full experience.

There's a bit of advice you didn't expect from this article — when you're headed to the movies, pick seats in the centre of the theatre.

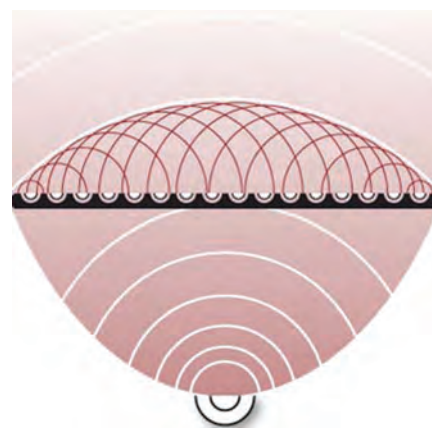
Obviously, if more of us are to enjoy our outings, we must all be in the sweet spot. One option, would have been to place speakers really far from the audience space. In an alternate universe, maybe they do have enough real estate to do that. In our world, we must find another way. It's called wave field synthesis.

The simulation

Imagine this: you're about ten feet away from a large speaker, looking (hypothetically) at the wavefront. This may be the optimal non-eyeball-melting-but-still-loud-enough distance, but what if you don't have ten feet to spare? What if you live in a high-rent dingy studio in the wrong part of the wrong metropolis, cursing your sad, pathetic...sorry, sometimes we get carried away. Anyway, when you're pressed for space, like say,

a new multiplex in a crowded metro, it is possible to use an array of smaller speakers and a mathematical algorithm or two to simulate that same wavefront, just inches away from you.

Creating the simulated wavefront



The array of speakers (in front) can create a wavefront that simulates the effect of a speaker far behind is simple enough. The speaker in the centre starts emitting sound first, the ones on either side next, and so on. And there you are — if the speakers simulate the wavefronts at the sweet spot, suddenly the sweet spot becomes every spot that's at least six inches away from